

Calculus I -- Midterm Examination

Spring 2026 -- 90 minutes -- 100 points

Name: *(write name here)* **Student ID:** *(write ID here)*

Instructions. Answer all five questions in the space provided. Show your work; partial credit is awarded for correct method even when the final number is wrong. Calculators are *not* permitted. A blank Answer Key appears at the end of the booklet for grader use only – students should not write in that section.

Question 1. Limits (15 points)

Evaluate the limit of $(x^2 - 4)$ divided by $(x - 2)$ as x approaches two. Show each algebraic step, especially the factoring that removes the indeterminate form.

Question 2. Differentiation (20 points)

Let $f(x)$ equal $x^3 \sin x$. Compute the derivative f' of x using the product rule, then evaluate f' at π .

Question 3. Chain rule (20 points)

Differentiate $g(x)$ equal to the square root of (one plus cosine of two- x) with respect to x . Simplify your answer until no double-angle identities remain.

Question 4. Definite integration (20 points)

Evaluate the definite integral, from zero to pi-over-two, of sine of x times cosine of x with respect to x . A u -substitution will make this routine – identify it explicitly before integrating.

Question 5. Application (25 points)

A spherical balloon is being inflated so that its radius increases at a constant rate of 2 cm/s. At the instant the radius reaches 10 cm, how fast is the volume increasing? Recall that the volume of a sphere is four-thirds πr^3 .

Answer Key

This page is for grader use only. Students should not write here.

Q1. Factor the numerator as $(x - 2)(x + 2)$. The factor of $(x - 2)$ cancels, leaving the limit of $(x + 2)$ as x approaches 2, which is 4.

Q2. By the product rule, $f\text{-prime}(x)$ equals $3x^2 \sin(x)$ plus $x^3 \cos(x)$. Evaluating at $x = \pi$ gives $f\text{-prime}(\pi) = 3\pi^2 \times 0$ plus $\pi^3 \times (-1) = -\pi^3$.

Q3. Let $u = 1 + \cos(2x)$, so $g = \sqrt{u}$. Then $g\text{-prime}(x) = (1 / (2\sqrt{u})) \times (-2 \sin(2x)) = -\sin(2x) / \sqrt{1 + \cos(2x)}$. Using $1 + \cos(2x) = 2 \cos^2(x)$ and $\sin(2x) = 2 \sin(x) \cos(x)$, this simplifies to $g\text{-prime}(x) = -\sqrt{2} \sin(x) \times \text{sgn}(\cos(x))$.

Q4. With $u = \sin(x)$, $du = \cos(x) dx$, the integral becomes the integral from 0 to 1 of $u du$, which is $1/2$.

Q5. Differentiating $V = (4/3) \pi r^3$ gives $dV/dt = 4 \pi r^2 dr/dt$. With $r = 10$ cm and $dr/dt = 2$ cm/s, $dV/dt = 800 \pi$, approximately 2513.3 cubic centimetres per second.